Martin-Luther-Universität Halle-Wittenberg
Faculty of Natural Science II
Institute of Physics

Physics Modelling Assistant

Using Agentic Artificial Intelligence in physics modelling

Bachelor’s Thesis
for the Award of the Academic Degree
Bachelor of Science
in Physik und Digitale Technologien

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1 introduction

Artificial Intelligence can be of great assistance in scientific research [35]. As shown by Chugunova et al. [5], researchers in natural science are among the groups most familiar with AI among scientists. There is a wide range of application stretching from finding sources in the literature to aiding in the presenting research findings and including the steps in between. It can help in forming hypothesis and in evaluating them. This includes the design and control of practical experiments as well as aiding during simulations especially with the program implementation.[35]

To improve the work with AI in scientific research automated AI research pipelines where developed in the last two years. The first fully autonomous approach was published by Lu et al. in 2024[23]. This framework identifies interesting research questions, plans and executes according experiments, interprets the results and writes a paper stating its findings in a multistep process.
The Less automated approaches where made by Shao et al. with the introduction of SciSciGPT [30]. This system is for Science of Science research which is why SciSciGPT is equipped with data management component amongst a literature review and an analytics component.

Use of AI in society and science often single models; reduces flexibility in tasks and multi step workflows

want to automate larger part of research so human can focus on important part: get model suggested from literature, see whether it's sufficient

=> multi step workflow

2 theoretical work

AI; LLM; AGENTICAI

LLM models

how agentic ai works

SAIA

Orchestration of Agentic ai

Crewai

Tools

3 Configurations of the pipeline

Task & Agent defs

Steps

Output

4 Benchmarking

Figure 1: testC

Table 1: testTab

a	b
c	d

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5 Appendix

6 Declaration of authorship

I hereby declare that I have written this thesis independently and without unauthorized assistance.

All sources and references used have been properly cited.

This thesis has not been submitted in the same or substantially similar form for any other degree or academic qualification.

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